

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Object Oriented Programming	Course Code:	CS1004
	Program:	BS Electrical Engineering	Semester:	Spring 2023
	Duration:	1 Hour	Total Marks:	40
	Exam Date:	11 th April, 2023	Weight:	15%
	Section:	All	Page(s):	2
	Exam Type:	Midterm-2		

- Instruction/Notes:**
- The exam is closed book and notes. Use valid assumptions if needed.
 - Please answer all questions on the answer book provided.
 - Use only black or blue pen to solve the paper.

Question # 1 A [CLO 2, 20 Marks]

A circle can be defined with a diameter/radius, whereas a cylinder has a height in addition to diameter/radius. Carefully read the following code, it defines and implements a class circle. You have to define and implement a class cylinder derived publically from class circle using the specifications provided after the code.

```
class circle{
private:
double diameter;
double radius;
double circumference;
double area;
public:
    void setDimensions( double d){
        diameter=d;
        radius=d/2;
        circumference=calCircumference();
        area=calArea();}
    void getDimensions( double& r , double& aRea, double& cirCumference){
        r= radius;
        aRea= area;
        cirCumference=circumference;}
    double calCircumference(){return 2*3.14159*radius;}
    double calArea(){return 3.14159*radius*radius;}
    void print(){
        cout<<" for circle object with diameter = "<<diameter<<endl;
        cout<<"radius = "<<radius<<endl;
        cout<<"circumference = "<<circumference<<endl;
        cout<<"area = "<<area<<endl;}
    circle(double d=0){setDimensions(d);}};
```

Following data attributes must be there in the derived class cylinder:
Height, cylindricalSurfaceArea, topPlusBottomArea, and Volume.
Following functions must be implemented

Overload setDimensions() function so that it receives two parameters a radius and a height.

Overload getDimensions() function so that it has an appropriate list of parameters

Constructor that receives two parameters a radius and a height.

Override/redefine print () function

Note: No cin/cout statement is allowed in class functions with a single exception as described in next sentence. You are allowed to use cout statements in print () function

Question # 2 [CLO 3, 20 Marks]

Overload + and stream operator << using friend functions for the class cylinder. Implement overloading using friend functions. Also write down a test program to check functionality implemented in Q1 and Q2.
